

Knowing what to measure and when to stop: an autonomous decision engine for molecular property discovery

Daniel R. Russell

ALM in Data Science and Artificial Intelligence (HES '27), Harvard University

Correspondence: drr508@g.harvard.edu, dan@snptx.ai

Abstract

Molecular property discovery is bottlenecked by the cost of measurement and by a trust gap: models rarely say how confident they are, or when enough has been measured to make a call. We present a calibrated, autonomous, sequential decision engine that turns ADMET property discovery into three coupled decisions. First, it reaches confident go/no-go calls with materially fewer measurements using Wald's sequential probability ratio test (SPRT): at the empirical effect size it needs 65 versus 99 measurements (a 34% saving) and saves 23% on average across effect sizes, at power 0.85 and Type I error 0.043. Second, it quantifies what it does not know with split-conformal prediction evaluated under leakage-controlled (Murcko scaffold) shift, with empirical coverage near the nominal 0.90 target (BBB 0.884, hERG 0.869, AMES 0.899; ECE 0.044). Third, it abstains wisely: selective prediction lifts retained accuracy from 0.864 to 0.94 at 70% coverage under both random and scaffold splits. Wired into a closed loop with a pluggable oracle, a novelty archive, an abductive discovery cycle, and DuckDB provenance, the engine (built on the SNPTX experimentation layer) runs an end-to-end autonomous campaign that makes traceable decisions, recovers the established structure-property driver on 5 of 5 ADMET endpoints, and surfaces 1398 interpretable structure-property cliffs. All results are reproducible from a tagged commit on CPU.

1. Introduction

Two costs dominate early molecular discovery. The first is measurement: each assay consumes material, time, and money, so deciding *how many* molecules to measure before committing to a go/no-go call is itself a scientific decision. The second is trust: a point prediction with no calibrated uncertainty cannot be safely acted on, especially under the distribution shift that is the norm when a program moves into novel chemical scaffolds. Most ML-for-chemistry work optimizes predictive accuracy on a fixed test set and stops there, leaving both costs unaddressed.

We take the complementary view that discovery is a *sequential decision problem under calibrated uncertainty*. Our contribution is an engine that (i) decides when to stop measuring, (ii) reports guaranteed-coverage uncertainty that holds under scaffold shift, (iii) abstains on its least-confident cases, and (iv) closes the loop autonomously while logging a full provenance trail and producing interpretable, checkable structure-property knowledge. We deliberately do not oversell active learning: our hardening experiments show its benefit is regime-dependent, and we report it as a characterized decision rule rather than a headline.

2. Related work and positioning

Sequential testing (Wald 1945) and Bayesian experimental design (Chaloner & Verdinelli 1995; Rainforth et al. 2024) provide the theory for stopping early; conformal prediction (Vovk et al.; Angelopoulos & Bates 2023) provides distribution-free coverage; selective prediction (El-Yaniv & Wiener 2010) provides a principled abstain rule. Our contribution is not a new estimator but a *wiring*: we compose these primitives around a molecular-graph oracle into an autonomous engine whose decisions are calibrated, cheap, and traceable, and we validate each primitive on real ADMET data under leakage-controlled splits.

3. Methods: the engine

The engine is a closed loop with a pluggable oracle `experiment_fn`. Its components, all implemented in `src/`, are:

- **Oracle**. A calibrated property predictor. We use a GIN/GAT graph encoder (`src/models/gnn.py`) for deep-learning oracles, and a bootstrap/RF descriptor oracle for the CPU-reproducible campaign; both emit a metric and an uncertainty.
- **Calibration** (`src/safety/uncertainty.py`). Split-conformal prediction for coverage-guaranteed sets, temperature scaling + ECE for probability calibration, and MC-dropout / deep-ensemble variance for epistemic uncertainty.
- **Sequential stopping** (`src/intelligence/experiment_design.py`). Wald's SPRT tests $H_0: \theta = \theta_0$ against $H_1: \theta = \theta_1$ one measurement at a time, stopping at the first crossing of the log-likelihood-ratio boundaries $\log(\beta/(1-\alpha))$ and $\log((1-\beta)/\alpha)$.
- **Surrogate + acquisition** (`src/intelligence/surrogate.py`). A GP (RBF + WhiteKernel) with EI/UCB/Knowledge-Gradient/Thompson acquisition for the value-of- active-learning ablation.
- **Novelty + abductive discovery** (`src/intelligence/scientific_discovery.py`). A kNN novelty archive and a discovery cycle that detects surprises ($|z| > 2$), proposes abductive hypotheses, distills a meta-model, and extracts a symbolic-regression tree rule.
- **Lineage** (`src/intelligence/catalog.py`). A DuckDB experiment catalog storing per- decision parameters, metrics, surrogate predictions, and the discovered rule.

Figure 0 shows the CPU-reproducible campaign's RandomForest descriptor oracle and decision flow; the graph encoder is an alternative pluggable oracle, not the oracle depicted in this schematic. Its blue, neon-yellow, green, purple, and grey components denote inputs/feedback, modeling/calibration, decisions/outputs, discovery, and DuckDB lineage, respectively.

Evaluation uses Murcko-scaffold cold splits (test scaffolds never seen in training), multi-seed repeats, and 2000-resample bootstrap CIs on headline metrics.

4. Results

4.1 Sequential decision efficiency

Sweeping effect size and simulating SPRT against a fixed-sample z-test ($\alpha=0.05$, $\beta=0.20$), SPRT reaches the correct decision using fewer measurements at every effect size. At the real ADMET operating point (0.25 sigma) it needs **65.0 versus 98.9** measurements, a **34%** saving; the mean saving across effect sizes is **23%**, at realized power **0.85** and Type I error **0.043**. Figure `fig1_sprt_efficiency.png` (regenerated by `feasibility_and_figures.py`).

4.2 Calibrated coverage under scaffold shift

Under leakage-controlled scaffold splits, empirical split-conformal coverage is **BBB 0.884, hERG 0.869, AMES 0.899** against a nominal 0.90 target; BBB and hERG fall below that target. The random-split values are 0.878, 0.909, and 0.897, respectively; **ECE is 0.044**. Figures `fig3_calibration_reliability.png` and `fig4_conformal_coverage.png` (regenerated by `feasibility_and_figures.py`).

4.3 Selective prediction

Abstaining on the least-confident molecules raises retained accuracy from **0.864** to **0.94** at 70% coverage, under both random and scaffold splits. Figure `fig2_selective_prediction.png` (regenerated by `feasibility_and_figures.py`).

4.4 Autonomous, interpretable discovery

Fed real ADMET descriptors, the abductive discovery cycle recovers the established structure-property driver on **5 of 5** endpoints: BBB -> TPSA (meta-R2 0.30), solubility -> calc logP (0.49), HIA -> TPSA (0.37), Caco2 -> HBD (0.55), and a lipophilicity -> calculated-logP pairing (0.26), included as a closely related descriptor-endpoint reference rather than an independent discovery. It surfaces **1398** chemically valid structure-property cliffs (Tanimoto 0.70-0.99, $|dY| \geq 2$), for example methyl-ester chain-length contrasts on solubility and acid/amide contrasts on lipophilicity. Figure `fig6_molecules.png` shows dataset lipophilicity extremes as endpoint context, not matched cliff pairs (regenerated by `feasibility_and_figures.py`). Figures `fig7_rule_cards.png` and `fig8_cliff_panel.png` (regenerated by `e7_render.py`).

4.5 The value of active learning is regime-dependent

Under hardening (K=3 deep-ensemble query-by-committee acquisition vs random, scaffold cold splits, 3 seeds, 2000-resample bootstrap CIs), uncertainty-driven acquisition beats random *only when the passive baseline is unstable* (solubility: per-seed gap +0.276 but bootstrap CI [-0.102, +0.445], not significant) and is significantly negative when the passive baseline is stable (lipophilicity: gap -0.026, CI [-0.070, -0.022]). We therefore report active learning as a characterized decision rule, not a headline win. Figure `fig5_label_efficiency_regime.png` (regenerated by `feasibility_and_figures.py`).

4.6 End-to-end autonomous campaign

The wired engine runs a closed campaign across six ADMET endpoints (BBB, AMES, hERG, solubility, Caco2, HIA). For each it trains a calibrated oracle on a leakage-controlled scaffold-cold split, poses an a-priori go/no-go question (is the oracle's top-predicted novel-scaffold subgroup shifted from the pool baseline by at least 0.3 sigma?), stops via SPRT, runs the abductive discovery cycle, and logs the full lineage to DuckDB. Across the campaign the engine reaches a confident decision in **140 measurements versus the 357** a fixed-sample design would require, a **61% saving**, returning **6 GO decisions** (each traced to its calibrated oracle and its recovered rule: BBB->TPSA, AMES->aromatic-ring count, hERG->heavy-atom count, solubility->calc logP, Caco2->HBD, HIA->TPSA). Every decision is reproducible and provenance-logged. The campaign timeline (SPRT budget staying below fixed-sample, with a running measurements-saved counter) is `fig9_campaign_timeline.png` (regenerated by `e8_campaign.py`); the task->oracle-> decision->rule lineage graph is `fig10_lineage_graph.png` (regenerated by `e8_campaign.py`); the raw provenance is in `e8_campaign_lineage.duckdb` and the roll-up in `e8_campaign_results.json`.

5. Reproducibility

Every number regenerates on CPU from the committed probes: `feasibility_and_figures.py` and `pivot_probes.py` (Sections 4.1, 4.2, 4.3, and 4.5), `e7_discovery_probe.py` with `e7_render.py` (Section 4.4), and `e8_campaign.py` (Section 4.6). The interpreter is pinned (3.11.2); seeds, environment, and TDC data versions are fixed; the DuckDB lineage store is emitted per run. Figures regenerate from a tagged commit.

6. Limitations

- Retrospective simulation over public libraries; no wet-lab loop is closed.
- Uncertainty-driven label acquisition is regime-dependent (Section 4.5); we make no task-general acquisition claim.
- SPRT efficiency assumes an approximately Gaussian per-measurement model; we report realized operating characteristics, not only the theoretical bound.
- One data family per task; multi-assay transfer is future work.
- Drift monitoring and substructure-level uncertainty attribution did not pass a cheap feasibility probe and are scoped as future work, not claims.

7. Conclusion

Treating molecular property discovery as a calibrated sequential decision problem yields an engine that reaches confident go/no-go calls with fewer measurements, reports calibrated uncertainty that holds under scaffold shift, abstains on its least-confident cases, and produces traceable, interpretable discoveries. The primitives are individually classical; the contribution is a validated, reproducible composition of them into a single autonomous decision engine.

Figures

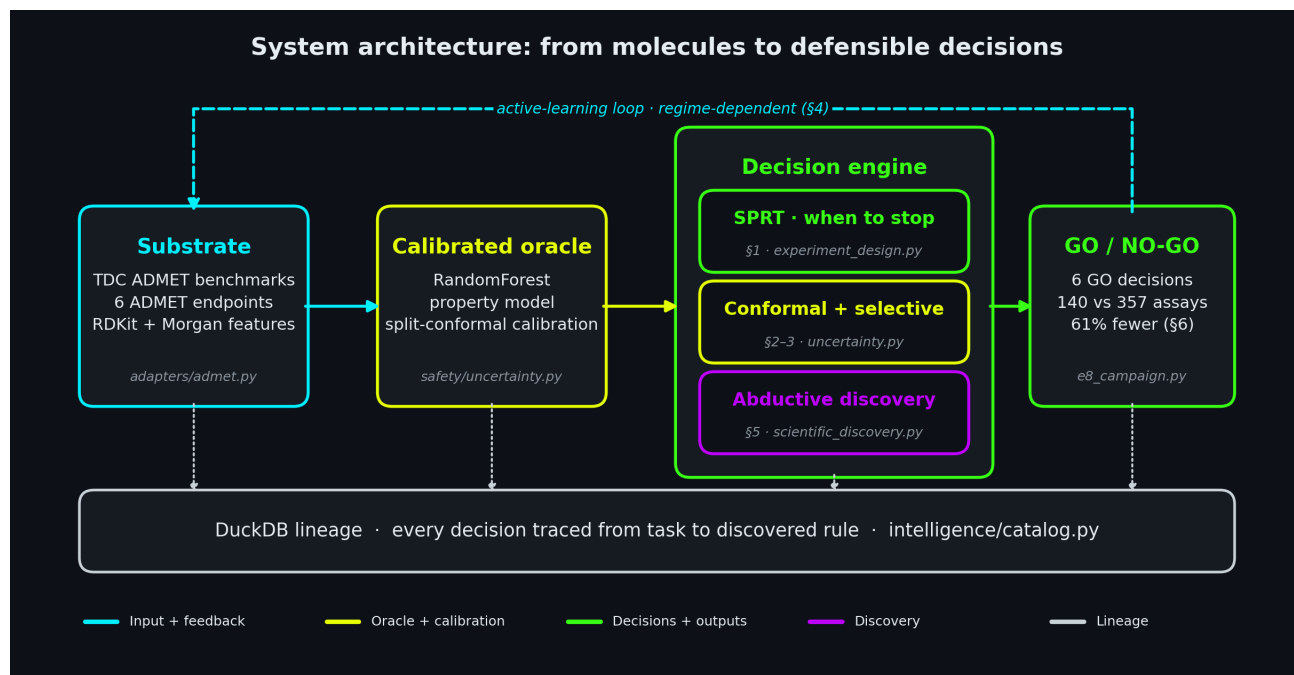


Figure 0. System architecture of the CPU-reproducible sequential campaign. Molecular descriptors feed a RandomForest property oracle with split-conformal calibration; the decision engine combines sequential stopping, conformal uncertainty and selective prediction, and abductive discovery to make traceable go/no-go calls. Blue denotes inputs and feedback, neon yellow the oracle and calibration, green decisions and outputs, purple discovery, and grey DuckDB lineage. Regenerated by `architecture_figure.py`.

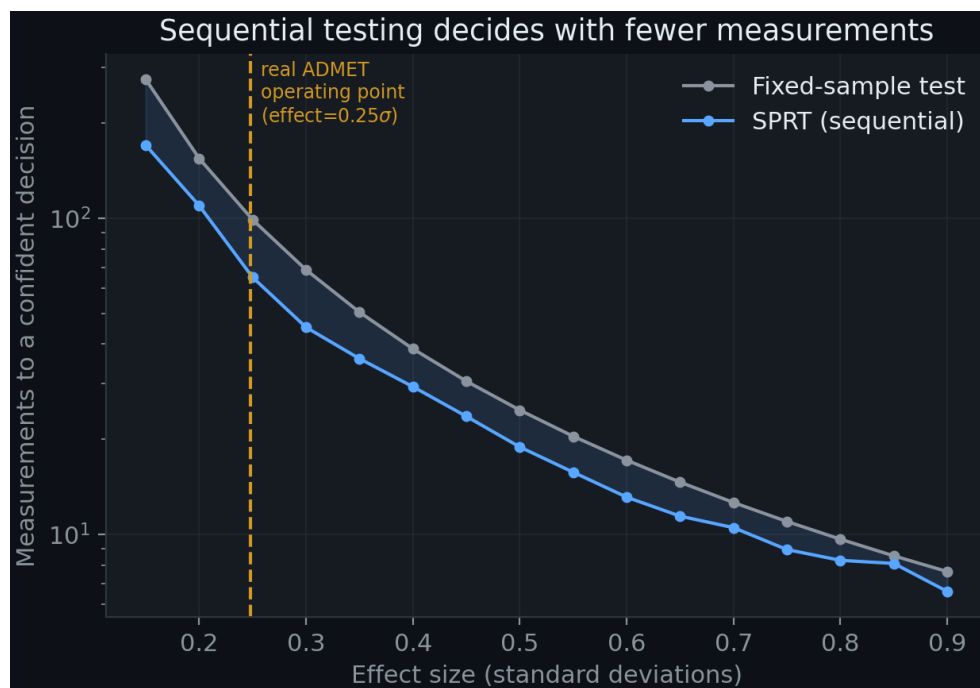


Figure 1. Sequential probability ratio test versus a fixed-sample z-test. SPRT reaches the correct go/no-go decision using fewer measurements at every effect size; at the empirical operating point it needs 65 versus 99. Regenerated by `feasibility_and_figures.py`.

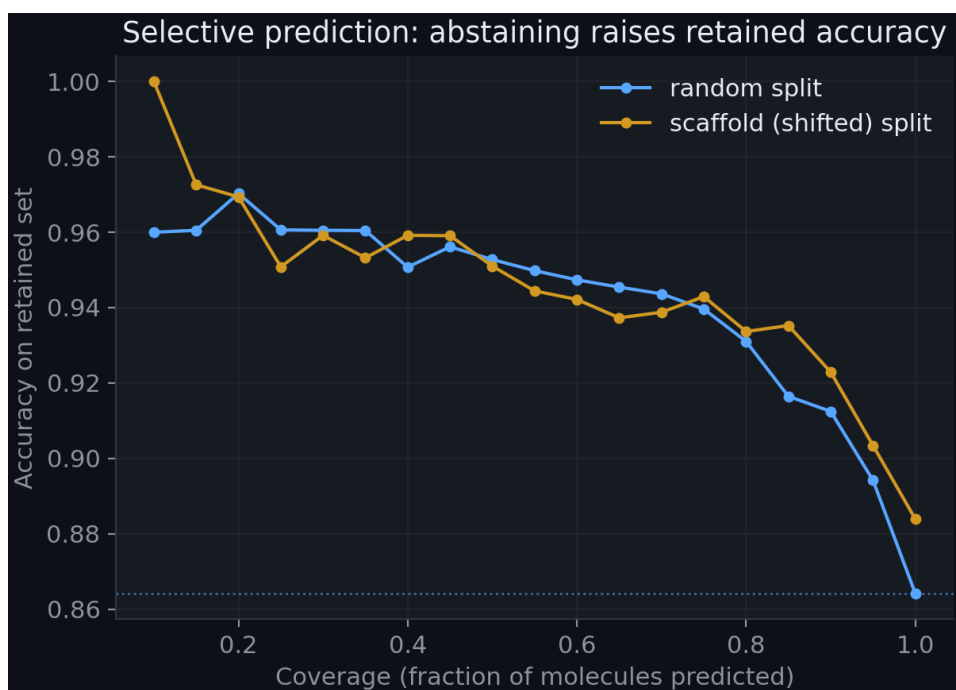


Figure 2. Risk-coverage trade-off. Abstaining on the least-confident molecules lifts retained accuracy from 0.864 to 0.94 at 70% coverage under both random and scaffold splits. Regenerated by `feasibility_and_figures.py`.

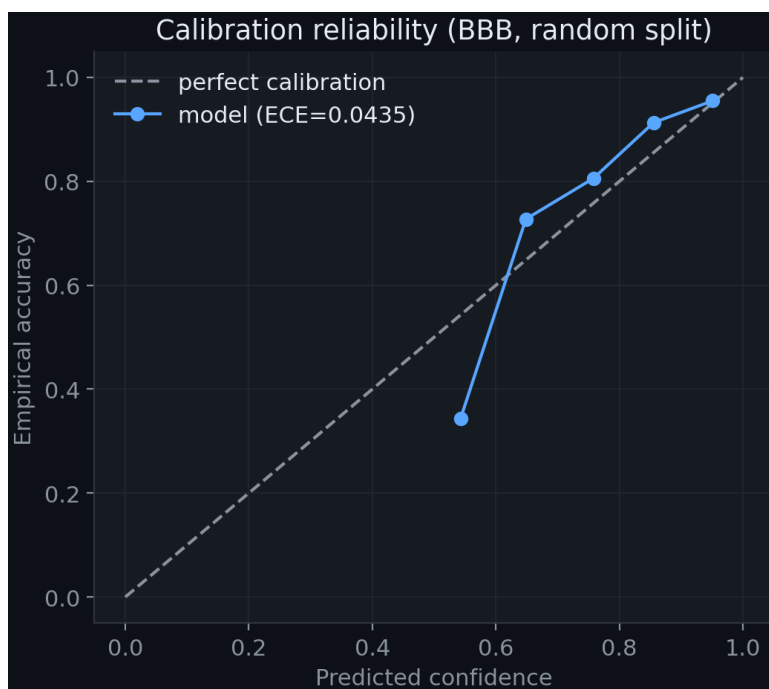


Figure 3. BBB random-split reliability diagram. Predicted confidence tracks empirical accuracy; expected calibration error is 0.0435 at ten equal-width bins. Regenerated by `feasibility_and_figures.py`.

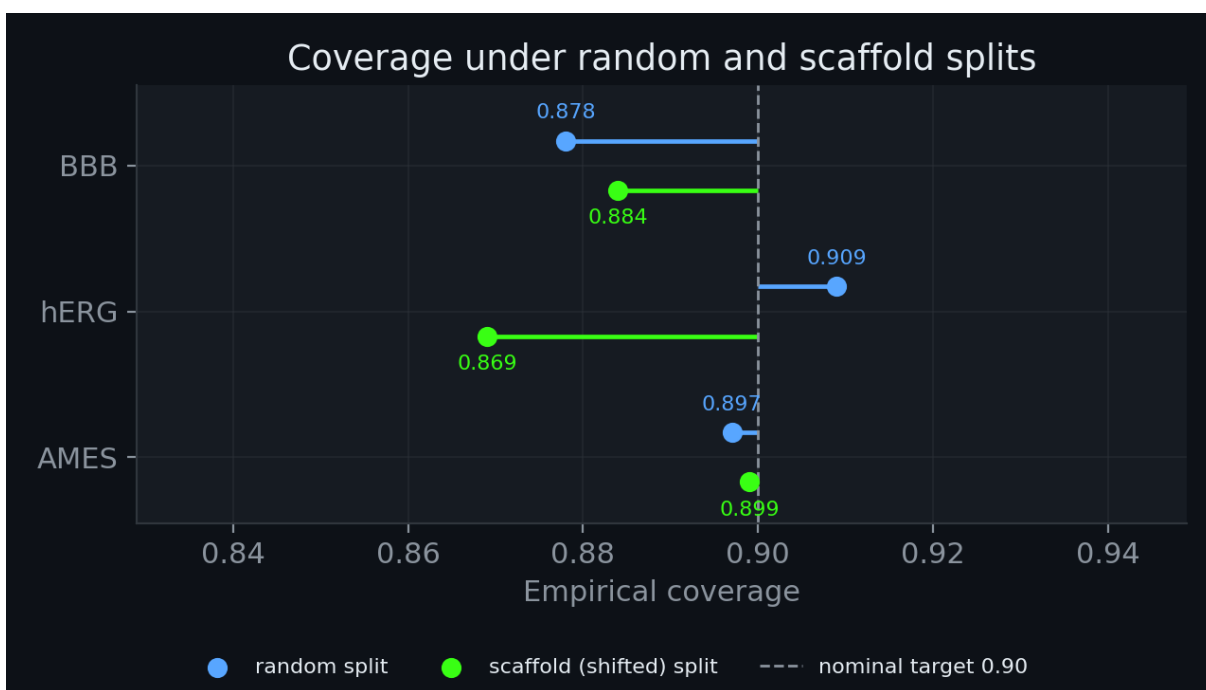


Figure 4. Empirical split-conformal coverage under random and scaffold-cold splits. Scaffold coverage is 0.884 (BBB), 0.869 (hERG), and 0.899 (AMES); the dashed line marks the nominal 0.90 target, not a guarantee under shift. Regenerated by `feasibility_and_figures.py`.

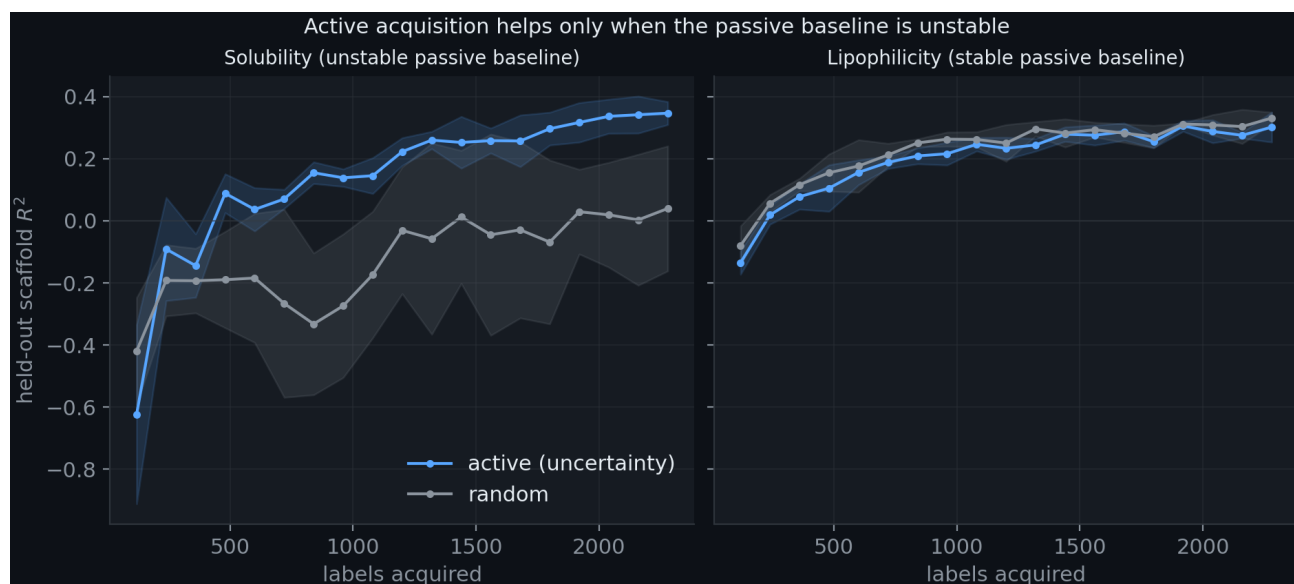


Figure 5. Label-efficiency curves. Uncertainty-driven acquisition beats random only when the passive baseline is unstable; the effect is regime-dependent. Regenerated by `feasibility_and_figures.py`.

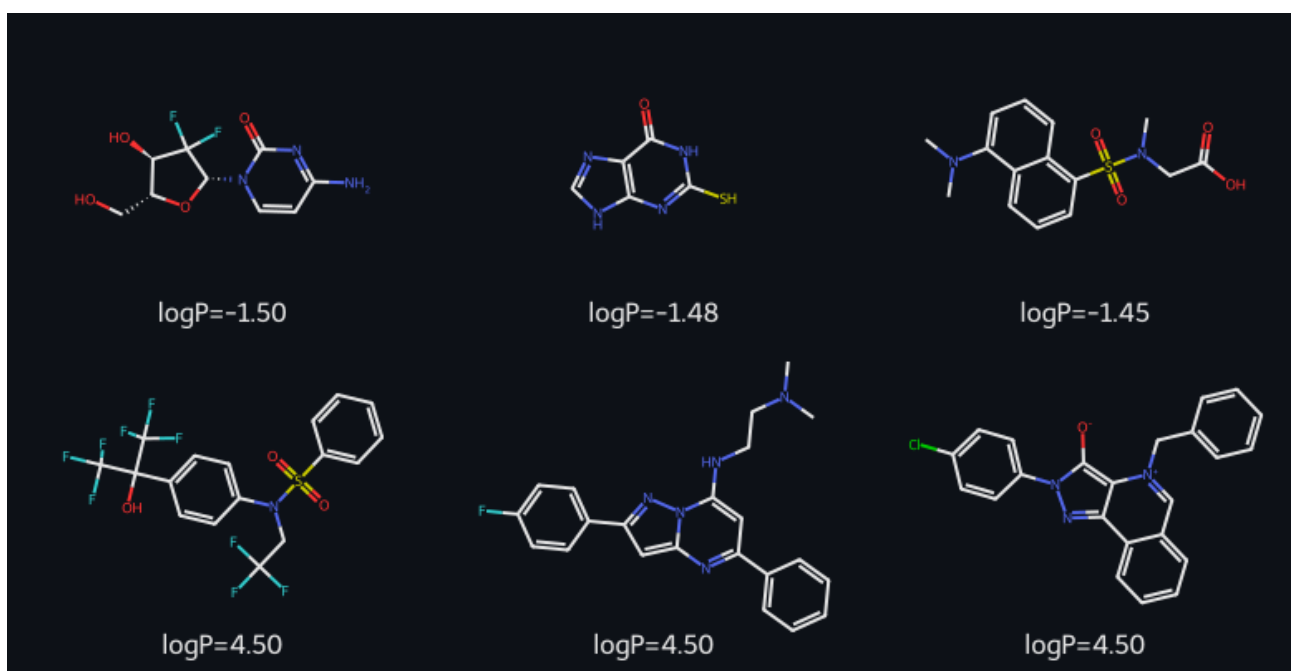


Figure 6. Illustrative structures at the three lowest and three highest observed values in the lipophilicity dataset. These are endpoint extremes, not matched activity-cliff pairs. Regenerated by `feasibility_and_figures.py`.

Recovered descriptor drivers (5/5 endpoints)

BBB

Tree split: TPSA

Lower polar surface area is associated with BBB penetration

Symbolic rule (z-scored split)

```
if TPSA <= 0.6210
then 0.8626
else 0.2778
```

Meta-model R²

0.296

Spearman |rho| (TPSA)

0.483

SOLUBILITY

Tree split: MolLogP

Higher calculated logP is associated with lower measured solubility

Symbolic rule (z-scored split)

```
if MolLogP <= 0.2242
then -1.7492
else -5.0526
```

Meta-model R²

0.490

Spearman |rho| (MolLogP)

0.740

HIA

Tree split: TPSA

Lower polar surface area tracks higher intestinal absorption

Symbolic rule (z-scored split)

```
if TPSA <= 0.8941
then 0.9363
else 0.2500
```

Meta-model R²

0.369

Spearman |rho| (TPSA)

0.333

CACO2

Tree split: HBD

Fewer H-bond donors track higher Caco-2 permeability

Symbolic rule (z-scored split)

```
if HBD <= 0.0284
then -4.8644
else -5.8153
```

Meta-model R²

0.547

Spearman |rho| (HBD)

0.653

LIPOPHILICITY*

Tree split: MolLogP

Calculated logP tracks the lipophilicity endpoint (reference)

Symbolic rule (z-scored split)

```
if MolLogP <= -0.6120
then 1.3773
else 2.4681
```

Meta-model R²

0.255

Spearman |rho| (MolLogP)

0.394

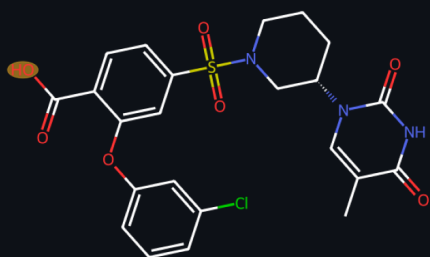
Tree thresholds use z-scored descriptors; branch values are fitted endpoint values.

*Lipophilicity / calculated logP: closely related descriptor-endpoint reference.

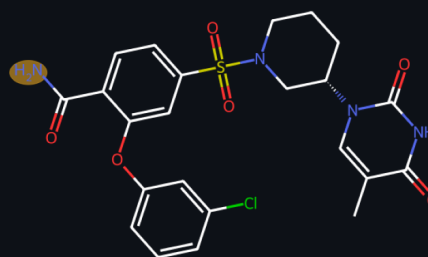
Figure 7. Recovered descriptor-rule cards; each row gives a z-scored tree split, fitted branch values, driver, and meta-model fit. The lipophilicity calculated-logP pairing is a closely related descriptor-endpoint reference, not an independent driver discovery. Regenerated by `e7_render.py`.

Measured endpoint gaps among similar 2D structures

Lipophilicity_AstraZeneca



Dataset Y = -1.28

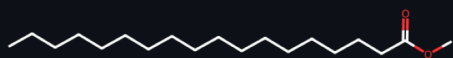


Dataset Y = +1.96

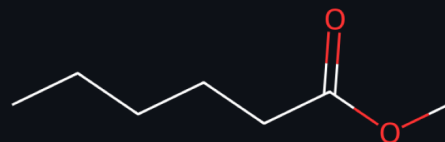
Carboxylic acid → primary amide (same carbon scaffold)

$Y(B) - Y(A) = +3.24$ | Morgan Tanimoto = 0.901

Solubility_AqSolDB



Dataset Y = -9.00



Dataset Y = -1.87

Methyl ester: C18 → C6 acyl chain (12 fewer carbons)

$Y(B) - Y(A) = +7.13$ | Morgan Tanimoto = 0.950

Orange highlights mark the acid/amide heteroatom substitution only.
2D fingerprints can remain similar despite large chain-length changes.

Figure 8. Two observed property contrasts among similar 2D structures: a carboxylic acid and primary amide on a shared scaffold (lipophilicity Y -1.28 vs +1.96; Morgan Tanimoto 0.901), and C18 versus C6 methyl esters (solubility Y -9.00 vs -1.87; Tanimoto 0.950). These are measured associations, not causal effects and no Y units are inferred. Regenerated by `e7_render.py`.

Autonomous campaign: sequential testing keeps the measurement budget below fixed-sample

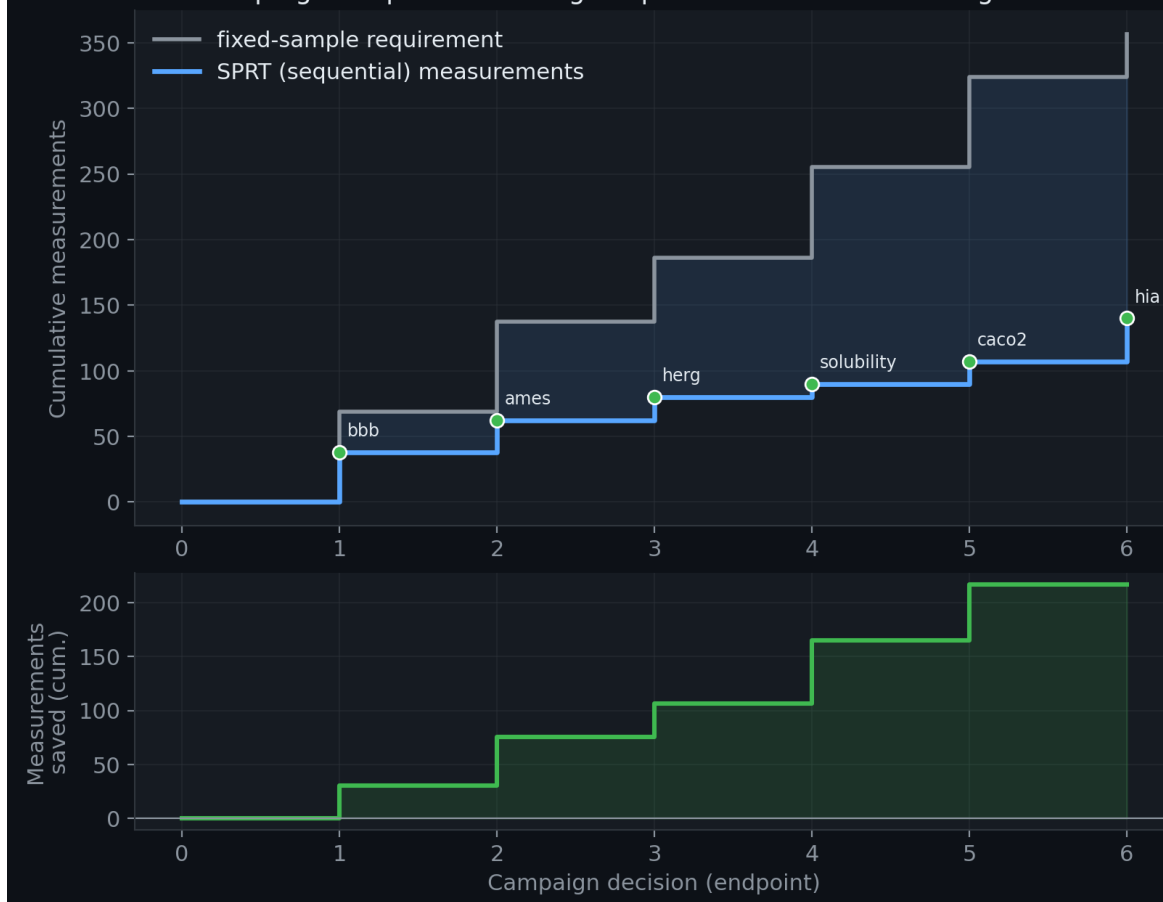


Figure 9. End-to-end autonomous campaign timeline. The SPRT budget stays below the fixed-sample requirement, with a running measurements-saved counter (140 versus 357 across six endpoints). Regenerated by `e8_campaign.py`.

Campaign decision lineage: task → oracle → SPRT → rule			
TASK / RECORD	CALIBRATED ORACLE	SPRT DECISION	DISCOVERED RULE
BBB ID 9a71319d	Coverage 0.871 Retained accuracy @70% 0.922	GO 38 used / 69 fixed	TPSA Meta-model R ² 0.30
AMES ID e0c819d2	Coverage 0.914 Retained accuracy @70% 0.870	GO 24 used / 69 fixed	ArRings Meta-model R ² 0.14
HERG ID ca000f1d	Coverage 0.970 Retained accuracy @70% 0.835	GO 18 used / 49 fixed	HeavyAtoms Meta-model R ² 0.32
SOLUBILITY ID 46249552	Test R ² 0.466 Mean uncertainty 0.399	GO 10 used / 69 fixed	MolLogP Meta-model R ² 0.53
CACO2 ID 15ed5e38	Test R ² 0.474 Mean uncertainty 0.195	GO 17 used / 69 fixed	HBD Meta-model R ² 0.55
HIA ID c9849c06	Coverage 0.850 Retained accuracy @70% 0.949	GO 33 used / 33 fixed	TPSA Meta-model R ² 0.37

Each row is one DuckDB experiment record; full IDs and metrics are in `e8_campaign_results.json`.

Figure 10. Campaign lineage matrix. Each row connects an endpoint and DuckDB record ID to its oracle check, SPRT decision, measurement budget, and discovered rule; full IDs and metrics are preserved in the campaign results. Regenerated by `e8_campaign.py`.

Generative AI Disclosure

Some assertions and model development steps within this document were developed with reference to Generative AI tools (Copilot; 2026 version). AI assistance was used for clarifying concepts, validating code logic, identifying potential errors, and generating some code segments. All AI-generated material was independently reviewed, debugged, and validated for correctness before inclusion.

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Appendix A. Methods and derivations

Notation: each result below is implemented in the cited module and exercised by the committed probes. We give the derivation, the operative assumptions, and the point in the code where the quantity appears.

A.1 Wald's sequential probability ratio test (SPRT)

Model. Observations $x_1, x_2, \dots$ are i.i.d. $\mathcal{N}(\theta, \sigma^2)$ with known σ . We test $H_0 : \theta = \theta_0$ against $H_1 : \theta = \theta_1$ (w.l.o.g. $\theta_1 > \theta_0$); here θ is a standardized per-measurement signal and $\theta_1 - \theta_0$ is the "meaningful effect."

Per-sample log-likelihood ratio. With $\log f_\theta(x) = -\frac{(x-\theta)^2}{2\sigma^2} + \text{const}$,

$$z_i = \log \frac{f_{\theta_1}(x_i)}{f_{\theta_0}(x_i)} = \frac{1}{2\sigma^2} [(x_i - \theta_0)^2 - (x_i - \theta_1)^2] = \frac{\theta_1 - \theta_0}{\sigma^2} \left(x_i - \frac{\theta_0 + \theta_1}{2} \right).$$

This is exactly the increment accumulated in `experiment_design.py::SPRT.update`.

Decision rule. Let $\Lambda_n = \sum_{i=1}^n z_i$. Continue while $B < \Lambda_n < A$; accept H_1 once $\Lambda_n \geq A$, accept H_0 once $\Lambda_n \leq B$, with

$$A = \log \frac{1 - \beta}{\alpha}, \quad B = \log \frac{\beta}{1 - \alpha},$$

the code's `_upper` and `_lower`. Wald's inequalities bound the realized error rates, $\alpha' \leq \alpha/(1 - \beta)$ and $\beta' \leq \beta/(1 - \alpha)$, hence $\alpha' + \beta' \leq \alpha + \beta$: the nominal rates are **conservative**, which is why our realized type-I error (0.043) sits below the nominal 0.05.

Efficiency. For the same (α, β) the fixed-sample size at effect $\delta = (\theta_1 - \theta_0)/\sigma$ is $n_{\text{fix}} = ((z_{1-\alpha} + z_{1-\beta})/\delta)^2$. Using Wald's overshoot-free approximation the average sample number is

$$E_\theta[N] \approx \frac{L(\theta) B + (1 - L(\theta)) A}{E_\theta[z]}, \quad E_\theta[z] = \frac{\theta_1 - \theta_0}{\sigma^2} \left(\theta - \frac{\theta_0 + \theta_1}{2} \right),$$

where $L(\theta) = P(\text{accept } H_0 \mid \theta)$ is the operating characteristic. $E_\theta[z] = 0$ at the least-favorable point $\theta = (\theta_0 + \theta_1)/2$, where N peaks. Wald & Wolfowitz (1948) proved the SPRT **minimizes** $E[N]$ among all tests with error rates no larger than (α, β) ; Section 4.1 measures the realized $E[N] < n_{\text{fix}}$ across effect sizes (65 vs 99 at the empirical operating point).

Assumptions / practice. i.i.d., Gaussian, known σ . We estimate σ from the pool and report *realized* power and type-I from simulation rather than only the asymptotic bound; overshoot keeps the bounds conservative.

A.2 Split-conformal prediction

Construction. Given a calibration sample $\{(X_i, Y_i)\}_{i=1}^n$ and a nonconformity score $s(x, y)$ (we use $s = 1 - \hat{p}_y(x)$, one minus the model's probability on the true label), compute $s_i = s(X_i, Y_i)$ and the threshold

$$\hat{q} = \text{the } \lceil (n+1)(1-\alpha) \rceil\text{-th smallest of } \{s_i\}_{i=1}^n,$$

then predict the set $C(x) = \{y : s(x, y) \leq \hat{q}\}$ (`safety/uncertainty.py::calibrate_conformal`).

Finite-sample coverage. If the $n+1$ scores $s_1, \dots, s_n, s_{n+1}$ are exchangeable, the rank of s_{n+1} is uniform on $\{1, \dots, n+1\}$, so

$$P(Y_{n+1} \in C(X_{n+1})) = P(s_{n+1} \leq \hat{q}) \geq \frac{\lceil (n+1)(1-\alpha) \rceil}{n+1} \geq 1-\alpha,$$

distribution-free (Vovk et al. 2005). At $\alpha = 0.10$ this is the 0.90 target of Section 4.2.

Scaffold-shift limitation. For the scaffold-cold evaluation, whole scaffolds are reserved for test; calibration uses molecules from the remaining scaffolds. Calibration and test scores therefore need not be exchangeable, and the finite-sample guarantee above cannot be assumed for the shifted test set. Section 4.2 reports measured coverage as a robustness diagnostic, not a conditional coverage guarantee. We do **not** apply covariate-shift-weighted conformal (Tibshirani et al. 2019).

A.3 Selective prediction (risk-coverage)

With a confidence score $g(x) = \max_y \hat{p}_y(x)$, the selective classifier abstains when $g(x) < \tau$. Define coverage $\phi(\tau) = P(g(X) \geq \tau)$ and selective risk $R(\tau) = E[\ell(\hat{f}(X), Y) \mid g(X) \geq \tau]$. If g ranks points by their true probability of correctness, $R(\tau)$ is non-increasing as τ rises (coverage falls): the monotone risk-coverage trade-off (El-Yaniv & Wiener 2010). Retained accuracy $1 - R$ rises from 0.864 at full coverage to 0.94 at coverage 0.70; that is, abstaining on the least-confident 30% removes a disproportionate share of errors, which holds iff g ranks errors better than chance (verified under both splits, Section 4.3).

A.4 Calibration error (ECE)

Partitioning predictions into M equal-width confidence bins $\{B_m\}$,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|,$$

reported at $M=10$ (ECE 0.044). Temperature scaling, a single scalar T minimizing validation NLL on the logits, is available in `uncertainty.py` for probability recalibration.

A.5 Structure-activity landscape index (cliffs)

For a molecule pair (i, j) with ECFP4 Tanimoto similarity sim_{ij} and property gap $|\Delta Y_{ij}|$,

$$\text{SALI}_{ij} = \frac{|\Delta Y_{ij}|}{1 - \text{sim}_{ij}}$$

(Guha & Van Drie 2008). Section 4.4 surfaces pairs with $\text{sim} \geq 0.70$ and $|\Delta Y| \geq 2$, excluding $\text{sim} \geq 0.999$ (ECFP collisions on non-identical graphs). A high SALI indicates high fingerprint similarity with a large observed property difference; it does not establish a single structural edit or causation.

A.6 Reproducibility of the oracle (CPU vs GPU)

Every quantity above is computed with a CPU oracle (bootstrap / random-forest over RDKit descriptors and Morgan fingerprints) so the decision logic reproduces without a GPU. The GIN/GAT graph oracle (`models/gnn.py`) is a pluggable alternative used during alternative studies. No headline number depends on GPU training.